Flood risk, regulated adaptation, and housing supply: Evidence from Florida

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Social vs. private values for flood safety

- Flood is the costliest form of natural disaster in the US.
  - Estimated $32 billion in damages per year.
  - Losses are expected to grow by 25% by 2050.
- Central concern: Wedge between social and private values for flood safety.
  - *ex-post* aid from Federal Emergency Management Agency (FEMA).
  - Low, weakly risk-differentiated premiums by the National Flood Insurance Program (NFIP).
- NFIP's primary tool for combating this inefficiency: Housing regulation.
  - Flood-safe building codes in the Special Flood Hazard Areas (SFHAs).
  - Approximately 10% of land nationwide.

Economic trade-offs: Correcting instrument vs. Distorting housing market

Trade-offs

- Floodplain regulation can work as a second-best instrument to correct inefficient exposure to flood risk.
- Widespread constraint on housing supply can be costly.
- Do Corrective benefits $>$ Distortion costs?

This paper ...

- Setting: 40-year time horizon of flood regulation in Florida.
- Estimates the the impact of long-run NFIP flood-plain regulation.
- Investigates the welfare impacts of current and counterfactual policies (using a model of residential choices and housing supply).

Preview of results

Quasi-experimental evidence (Spatial discontinuity & event study design)

- By 2016, the share of developed land is 16% lower just inside the regulated area (31% difference in new construction).
- Such effect is not accompanied by a large reduction in price (less than 1%).
- Building standards reduce average flood damage by 56%.
- Homeowners do not privately value this reduction in flood risk.

Empirical model of residential demand and housing supply

- Expected flood damage 61% $\downarrow$ ($ 3.3B per county); House price 2.9% $\uparrow$ market-wide.
- *Status quo* achieves close to the efficent level of damage reductiosn (90%) but does not do so efficiently (30% of the social welfare gains of the actuarially-fair pricing benchmark).

Related literature

1. **Evaluation of natural disaster mitigation policies.** Hino and Burke (2021); Beltrán et al. (2018); Fried (2022); Baylis, Boomhower, et al. (2026).
2. **Frictions in mitigating or adapting to climate risk.** Gallagher (2014); Annan and Schlenker (2015); Deryugina and Kirwan (2018); Kousky et al. (2018); Bakkensen and Barrage (2022); Wagner (2022); Baylis and Boomhower (2023); Mulder (2024); Balboni (2025); Hsiao (2025).
3. **Effect of second-best corrective policy on reducing social welfare losses.** Ito and Sallee (2018); Jacobsen, Knittel, et al. (2020); Kellogg (2020); Barahona et al. (2023); Brot-Goldberg et al. (2023); Jacobsen, Sallee, et al. (2023).

NFIP and flood mapping process

NFIP (est. 1968)

- Publishes Flood Insurance Rate Maps (FIRMs) delineating areas by flood risk.
- Designates high-risk areas as Special Flood Hazard Areas (SFHAs), where new construction and substantial improvements must meet NFIP standards.

SFHA mapping process

- Due to engineering limitation, flood map delineation is imprecise on the boundary (especially for flat lands like Florida).
- In principle, FEMA maps are updated every 5 years.
- In practice, updates are rare (50% of maps were more than 5 years old as of 2017).
- Maps are highly persistent over time (80% of land in the sample is classified consistently over the approximately 40-year period).

Building codes



Figure: Flood safe house in Florida

Data I

Digitized historic and current flood maps

- A set of archival scans of early flood maps for parts of eleven counties in Florida.
  Digitized map

Land use data

- Spatially-granular data on land use to study outcomes on either side of the boundary.
- Use satellite-derived land use data to measure historical and current "development."

Parcel characteristics

- Florida Department of Revenue property tax records (2005-2020).
- Housing characteristics including sales prices, parcel outlines, and location.

Data II

Flood risk

- NFIP claims and policies (2010-2018) to measure flood damages for insured structures.
- Spatially-granular estimates of flood risk from a third-party hydrological model produced by First Street.

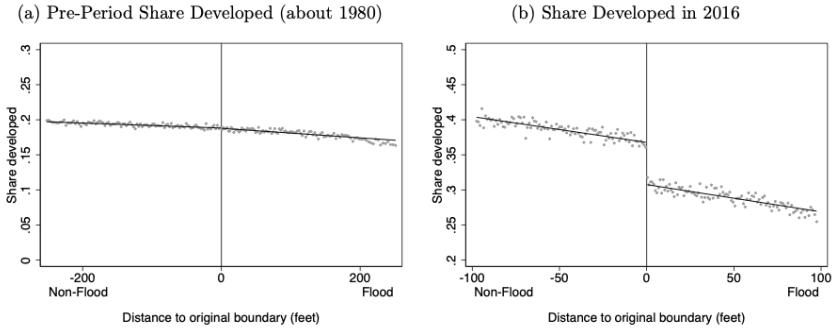
Spatial regression discontinuity around SFHA boundaries

$$y_p = \beta 1\{d_p > 0\} + f(d_p) + \mu \cdot \mathbf{x}_p + \varepsilon_p, \tag{1}$$

where:

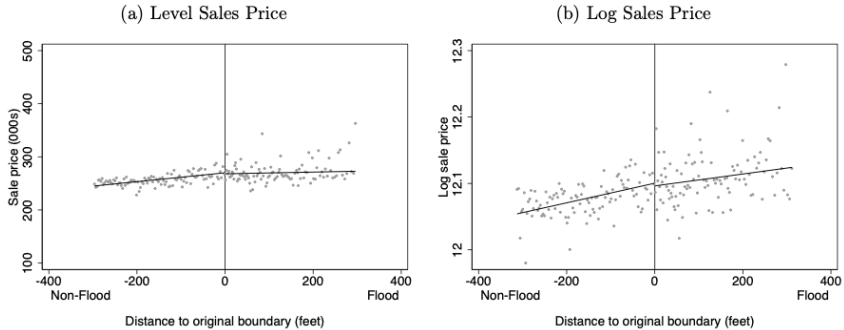
- y_p is a characteristic of pixel p;
- d_p is the distance from each pixel to the nearest flood zone boundary (positive $=$ inside the flood zone);
- Control for local linear functions $f(d_p)$ that are allowed to differ on either side of the boundary.
- The vector $\mathbf{x}_p$ includes boundary segment controls.
- For hedonic specifications with price, the equation is in transaction-level and the equation controls for transaction and parcel characteristics.

Results: Development



- Land just inside the flood zone is 6 percentage points less likely to be developed.
- 16% of the outside-flood-zone mean level of development and a 31% reduction as a share of new development (between 1980-2016).

Results: Prices



- Summary: Quantity of housing (supply side) falls significantly, despite modest price reductions.

Event study around the introduction of building standards

$$m_{kct} = \sum_r 1\{c - e_k = r\} + \gamma_k + \gamma_t + \varepsilon_{kct} \tag{2}$$

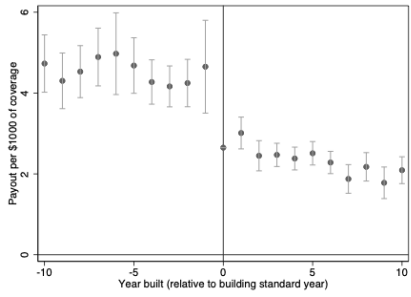
$$p_{js} = \sum_r \beta_r 1\{c_j - e_{k(j)} = r\} + \gamma_{k(j)} + \mu \cdot \mathbf{x}_{js} + \varepsilon_{js},$$

where:

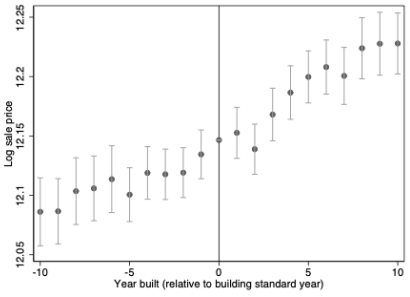
- m_{kct}: insurance payouts/premiums per dollar of coverage in policy year t for homes in census tract k constructed in year c.
- p_{js}: log sales price of house j sold in transaction s.
- r is the construction year relative to the year e_k of NFIP enrollment of the census tract.
- $\mathbf{x}_{js}$ are sale and parcel characteristics including year-by-month of sale fixed effects.

Results: Flood damages and prices

(a) Damages (Insurance Payouts)

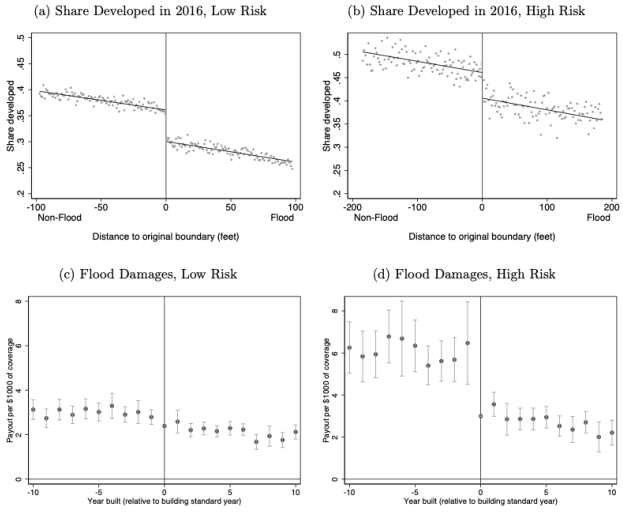


(b) Log Sale Price



- Land just inside the flood zone is 6 percentage points less likely to be developed.
- 16% of the outside-flood-zone mean level of development and a 31% reduction as a share of new development (between 1980-2016).

Heterogeneity by flood risk



The empirical model of the housing market: Residential demand

- Each consumer i makes a discrete choice of where to live among the houses j (housing types h) within a market (county) m.

Following Bayer et al. (2007), the indirect utility of consumer i choosing a house j is:

$$u_{ij} = p_{h(j)} \cdot \alpha(\mathbf{z}_i) + SFHA_{h(j)} \left(\phi + \rho Post_{h(j)}\right) + \mathbf{x}_{h(j)} \cdot \pi(\mathbf{z}_i) + \theta_{n(j)b(j)} + \upsilon_{r(j)} + \xi_{h(j)} + \varepsilon_{ij}. \quad (3)$$

Assuming $\varepsilon_{ij} \overset{\text{i.i.d.}}{\sim}$ T1EV

$$P_j(\mathbf{z}_i) = \frac{\exp\left(\delta_{h(j)} + \lambda_{h(j)}(\mathbf{z}_i)\right)}{\sum_{j' \in J_m} \exp\left(\delta_{h(j')} + \lambda_{h(j')}(\mathbf{z}_i)\right)}. \quad (4)$$

- This revealed preferences approach estimates willingness to pay for SFHA status $\left(\frac{\phi}{|\alpha(\mathbf{z}_i)|}\right)$, and adaptation $\left(\frac{\rho}{|\alpha(\mathbf{z}_i)|}\right)$.

The empirical model of the housing market: Housing supply

- Housing types defined by neighborhood n, flood zone status z, and boundary proximity indicator b.
- Given: the number of pre-standards houses as of 1980 (N_{nzb}^{1980}).

Following Baum-Snow and Han (2024), model development with a constant-elasticity supply curve:

$$N_{nzb}^{2016} = N_{nzb}^{1980} \exp(\psi \overbrace{E_z}^{\text{flood zone indicator}} + \mu_{nb} + \underbrace{\eta_{nzb}}_{\text{idiosyncratic location-specific construction cost shifters}}) \left(\frac{p_{nzb}^{2016}}{p_n^{1980}} \right)^{\gamma_n} \tag{5}$$

- ψ: Regulatory supply cost.

Model estimation: Demand

Similar to BLP

- Outer layer: Given $\lambda_h(\mathbf{z}_i)$, solve for δ_h using contraction mapping.
- Inner layer: Given δ_h, run regression with following assumptions justified through previous quasi-experimental evidence.
  - $E[SFHA_h 1_b \xi_h] = 0$: Conditional on boundary FE, and within 100 feet of the boundary, ξ_h and $SFHA_h$ are uncorrelated.
  - $E[SFHA_h Post_h 1_w \xi_h] = 0$: Within narrow 2-year vintage window, ξ_h and $Post_h$ are uncorrelated.
  - Instrument price with the slope of the land underneath house type h (shifts the cost to build).

Model estimation: Supply

- Challenge: Regulatory supply cost ψ can be correlated with unobserved construction costs η_{nzb}.
- Assumption: Within 100 feet of the flood zone boundary, unobserved construction costs are uncorrelated with flood zone status: $E[SFHA_z 1_b \eta_{nzb}] = 0$.
- We run following rearranged regression:

$$(\log N_{nzb}^{2016} - \log N_{nzb}^{1980}) - \underbrace{\gamma_n}_{\text{Baum-Snow and Han (2024)'s supply elasticities}} (\log p_{nzb}^{2016} - \log p_{nzb}^{1980}) = \psi SFHA_z 1_b + \zeta SFHA_z (1 - 1_b) + \mu_{nb} + \eta_{nzb}. \quad (6)$$

Parameter estimates (brief)

Demand

- Price elasticity (average): ≈ -1.9.
- Modest distaste for living in the flood zone ($-\$6,900$); Preference for adapted homes ($\$4,900$).

Supply

- ψ: -0.107 ($\approx$ increase in costs of approximately 14%).
- Corresponds to substantial average cost of compliance ($\approx \$18,000$).

Model-informed estimates: Impact/mechanism of NFIP policy on real estate development

Counterfactual benchmark

- Unregulated policy regime (flood zone status $SFHA_h = 0$, adaptation mandate $E_h = 0$, and taxes $\tau_h = 0$).

status quo vs. unregulated regime

- 900,000 houses are adapted (22% of all new development)
- 125,000 houses are reallocated out of the regulated floodplain (12% reduction in construction in regulated areas).
- Expected damages decrease by $\$8,141$ per newly-developed house (61% reduction relative to unregulated benchmark).
- Effect is mostly driven from supply-side housing regulation.

Model-informed estimates: Does the program encourage excessive flood risk compared to an actuarially-fair insurance scheme?

Counterfactual benchmark

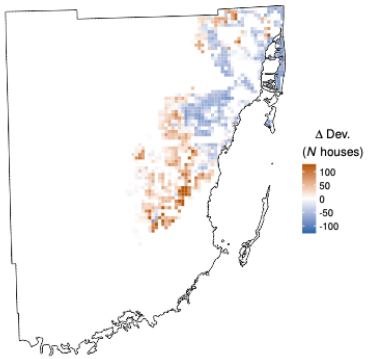
- Actuarially-fair insurance pricing regime (flood zone status $SFHA_h = 0$ and consumers face a tax equal to expected flood damages, varying with both location and adaptation status).

status quo vs. actuarially-fair pricing regime

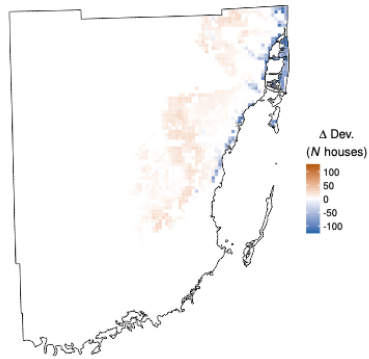
- *status quo* policy achieves almost all (87%) of the damage reductions under actuarially-fair pricing.
- This difference is entirely due to the influence of the NFIP's supply-side housing regulations.

But status quo policy might be too expansive!

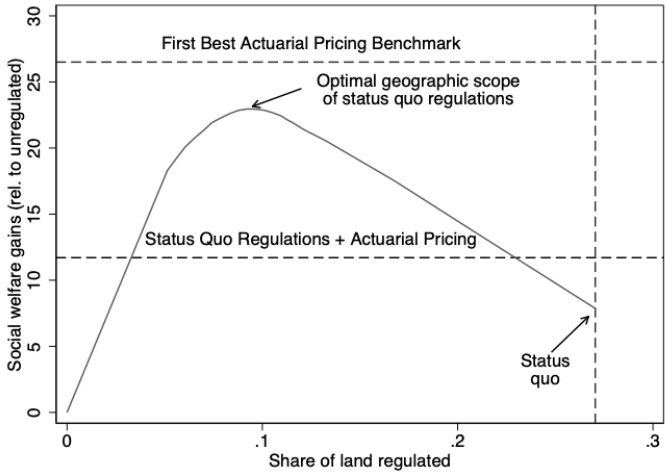
(a) *Status Quo* Policy (vs. Unregulated)



(b) Actuarial Pricing (vs. Unregulated)



Model-informed estimates: Do the flood-risk reductions balance the costs imposed through restrictions on housing supply?



Conclusion

- This paper investigates the impact of long-run (40 years) floodplain regulations on U.S. housing markets.
- Using regression discontinuity and event study design:
  - Regulation decreases construction but has negligible effects on house prices.
  - Flood-safe building standards reduce flood damages, but these reductions are not privately valued by consumers.
- Using empirical model of residential choice and housing supply:
  - Demand-side margin (residential choices) is small.
  - *Status quo* policy's approximately efficient levels of flood risk is due to its strong supply-side adaptation requirements.
  - Due to *Status quo* policy's too expansive regulation, it implements lower social welfare gains than less-expansive regime.

Thank you!

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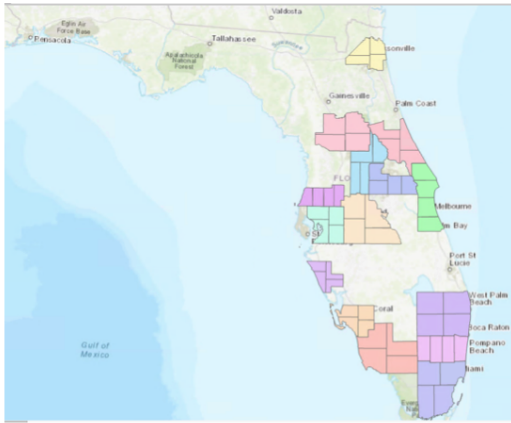
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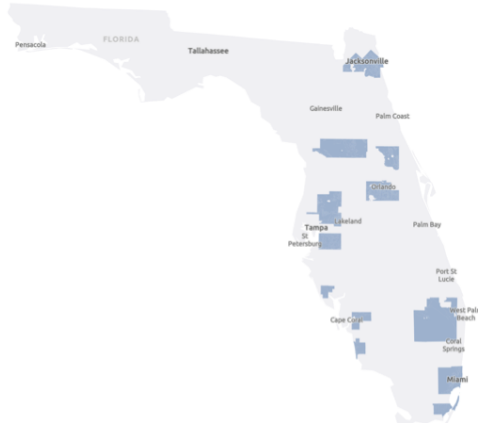
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Appendix

(a) Areas Considered for Digitization



(b) Coverage of Digitized Maps



Back to slide